Implementing Verdier's robust extrapolation in the CKT model

Relaxing the i.i.d. assumption on ν_{it}^{l}

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Our current estimator is Verdier's "simple" extrapolation

We currently follow Suri (2011) and Tjernström (2023):

- impose the LCA restriction $\Delta_i = \beta + \phi\,\theta_i$
- identify ϕ from switcher trajectories
- extrapolate to non-migrants (d_N, d_T) along the LCA line

Verdier calls our approach the *simple extrapolation* & proposes a *robust* alternative

The simple extrapolation assumes i.i.d. non-pecuniary shocks

Our current Assumption A2 states that the non-pecuniary utility shock

$$\nu_{it}^{l} \quad \text{is i.i.d. across individuals } i, \text{ time } t, \text{ \& locations } l$$

This implies:

- ν_{it}^{l} independent of baseline skill & comparative advantage everywhere
- rules out systematic differences in ν_{it}^{l} across regions/institutions
- workers in different locations have the same shock distribution, conditional on comparative advantage

The migration literature documents what A2 rules out

Regional heterogeneity is a central fact of this literature:

- **Bryan & Morten (2019):** structural model of Indonesian internal migration estimates large bilateral migration costs
- **Lagakos et al. (2020):** migration cost estimates differ substantially across developing countries
- **Pulido & Swiecki (2021):** cross-sector barriers in Indonesia explain a large share of within-country income gaps
- **China:** hukou is an institutional source of region-specific migration barriers

Our current model rules this out by assumption

Verdier's robust extrapolation weakens A2 to a conditional version

Decompose the non-pecuniary shock:

$$\nu_{it}^{l} = m_l(v_i) + \tilde{\nu}_{it}^{l}$$

where v_i is an observed, time-invariant individual-level index

New assumption (A2′):

- $\tilde{\nu}_{it}^{l}$ is i.i.d. *conditional on* v_i
- $m_l(v_i)$ can correlate arbitrarily with comparative advantage

We use first-wave province as v_i to index region-level shocks (all three countries)

Why this is weaker

A2 required *unconditional* independence

A2′ requires only *within-province* independence

The LCA gets a province-specific intercept

Slope stays common; intercept varies

Simple:

$$\Delta_i = \beta + \phi\,\theta_i$$

Robust:

$$\Delta_i = \beta(v_i) + \phi\,\theta_i$$

- slope ϕ remains common—this is still the LCA restriction
- intercept $\beta(v_i)$ absorbs province-level $m_l(v_i)$ heterogeneity
- ϕ identifies off within-province switcher variation
- extrapolation to non-migrants happens province-by-province, then averaged

Pooling provinces mixes differentially-selected switcher samples

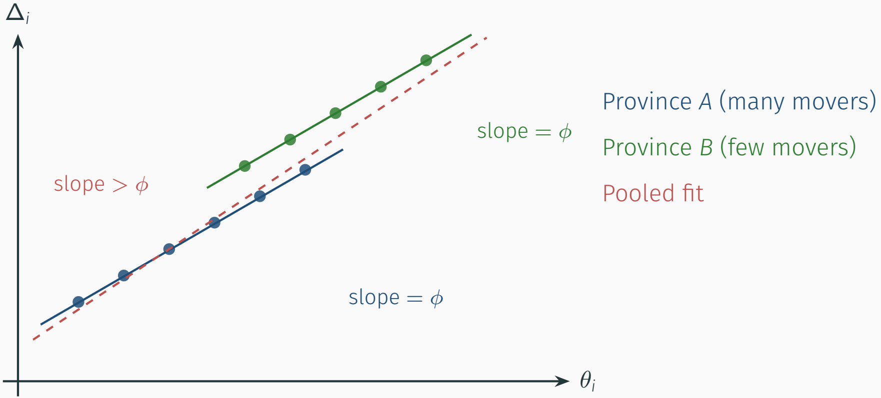
Consider two provinces:

- **Province A** (many movers): wide switcher sample, broad range of θ_i
- **Province B** (few movers): skewed sample, only high-return types migrate

Pooled regression: biased by across-province difference in selection

Within-province: clean slope, holds region-level shocks constant

Within-province slopes agree—the pooled slope is pulled off



Each province's switcher cloud has true slope ϕ

The pooled line tilts because of intercept differences, not because ϕ varies

Implementation adds v_i-interacted means to the GRC regression

Currently:

$$y_{it} = \sum_{\underline{d}} \mu_{\underline{d}} \mathbb{1}\{\underline{d}_i = \underline{d}\} + \Delta_{\underline{d}_0} D_{it} + \sum_{\underline{d} \in \mathcal{D}_S \setminus \underline{d}_0} \phi(\mu_{\underline{d}} - \mu_{\underline{d}_0}) D_{it} \mathbb{1}\{\underline{d}_i = \underline{d}\} + \cdots + \varepsilon_{it}$$

Under robust: trajectory means $\rightarrow$ trajectory $\times$ province means

$$y_{it} = \sum_{\underline{d},v} \mu_{\underline{d},v} \mathbb{1}\{\underline{d}_i = \underline{d}, v_i = v\} + \Delta_{\underline{d}_0} D_{it} + \phi \sum_{\underline{d} \neq \underline{d}_0} (\mu_{\underline{d},v_i} - \mu_{\underline{d}_0,v_i}) D_{it} \mathbb{1}\{\cdot\} + \cdots$$

- slope ϕ unchanged—still a single scalar
- standard errors cluster at v_i
- stayer return Δ_{d_N} averaged over clusters with switcher support

The move buys a weaker assumption at tractable cost

We gain:

- A2′ matches what the migration literature documents
- the CHN hukou split becomes a principled special case (v_i = hukou) rather than an ad hoc fix
- stayer ATE estimates defensible against Verdier's concern in Suri's data

We give up:

- Δ_{d_N} averages over provinces with switcher support, not the full population
- thinner identification of ϕ: only within-province switcher variation contributes
- cluster-robust standard errors at the v_i level

An observed shifter w_i provides a separate falsification test

Regress estimated θ_i^* on Δ_i and an observed shifter w_i:

$$\theta_i^* = \alpha_0 + \alpha_1 \Delta_i + \alpha_2 w_i + e_i, \qquad H_0 : \alpha_2 = 0$$

Kenya: $w_i =$ distance to nearest seed seller; rejects if $\alpha_2 \neq 0$

w_i must be observed, exogenous to skill conditional on Δ_i, and correlated with ν_{it}^l

Candidates in our data:

- distance / travel time to nearest urban center
- rainfall variability or pre-period climate shock (mergeable via GPS/admin code)
- distance to family already in urban area (closest analogue, but reflection problem)

None is perfect; data availability is the binding constraint